# Formation of the Solar System

**Exam code:** 0625 & 0972

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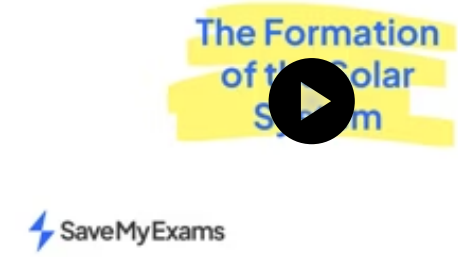


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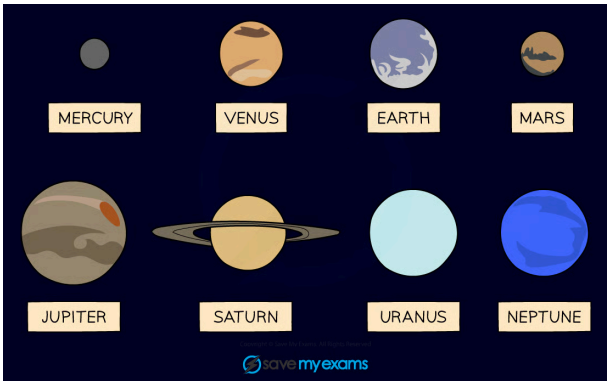
Help me understand

## Formation of the Solar System

- The 4 **inner** planets (nearest to the Sun):
  - are **rocky** and **small**
  - have **atmospheres** (except for Mercury)
- The 4 **outer** planets (furthest from the Sun):
  - are **gaseous** and **large**
  - are mostly composed of **hydrogen** and **helium** gas

### The planets in the Solar System




**Stuck on something?**

Ask questions and get help with exactly what you're revising.

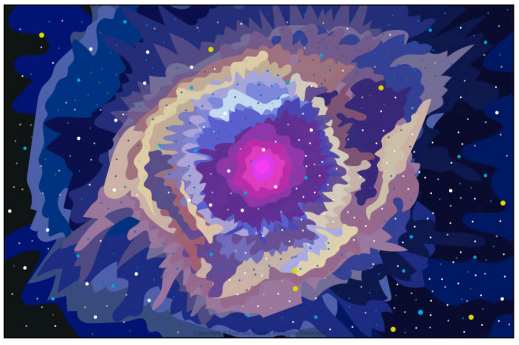
Got it

- The differences between the inner and outer planets can be explained using the **accretion model** for the formation of the Solar System

## Distribution of elements in the Solar System

- The Sun and the planets in the Solar System formed from a **cloud of dust and gas** (nebula)
  - Gravity pulled this cloud together into a giant ball, which would eventually become the Sun
  - As the nebula collapsed, the Sun became **denser** and **hotter**
- The Solar System then formed around 4.5 billion years ago
  - The planets formed from the remnants of the matter left over from the nebula that formed the Sun
  - The nebula contained many elements that were created during a supernova explosion in the distant past
- As the early Sun became hotter, gaseous matter was pushed **further** out into the Solar System than solid matter

### A nebula



***A nebula is an interstellar cloud of gas and dust***

### Formation of the inner planets

- In the hotter regions, closer to the Sun, the temperature was too high for lighter elements to exist in a solid state
  - Therefore, the inner planets formed from materials with high melting temperatures such as metals (e.g. iron)
  - The original nebula contained only a small proportion of heavy elements, so the inner planets could not grow as significantly as the outer planets
- As a result, solids in the inner disc were pulled together by gravity to form solid planets
  - This is why the 4 planets nearest to the Sun (Mercury, Venus, Earth and Mars) are **rocky and small**

### Formation of the outer planets

- In the cooler regions, further from the Sun, the temperature was low enough for the light molecules to exist in a solid state

    - The original nebula contained a large proportion of light elements, so the outer planets were able to become exceptionally large

- As a result, gases in the outer disc were pulled together by gravity to form gaseous planets
    - This is why the 4 planets furthest from the Sun (Jupiter, Saturn, Uranus and Neptune) are **gaseous and large**

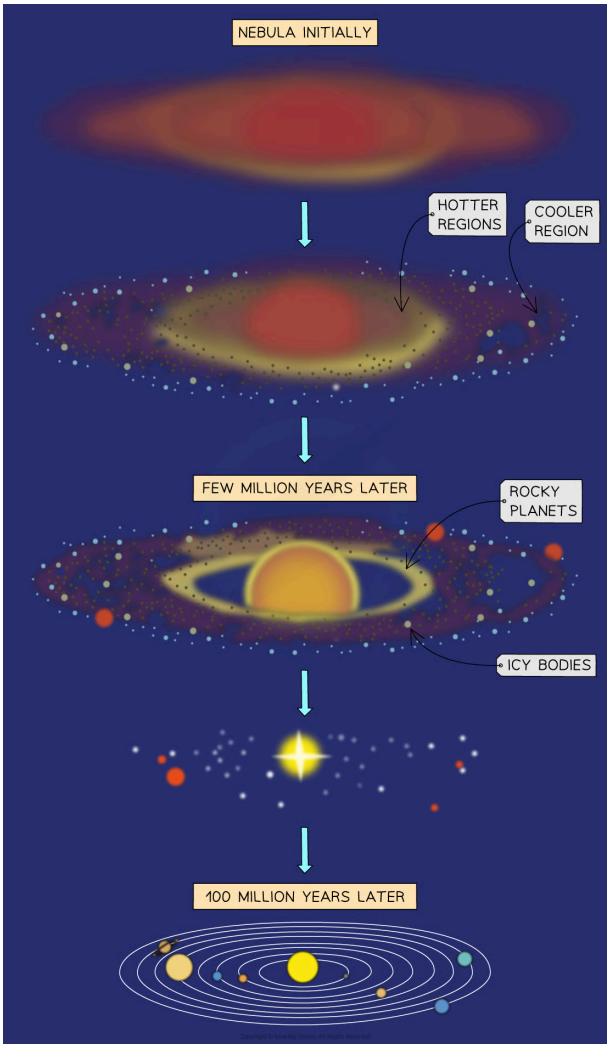
## Formation of the accretion disc

- In the nebula, matter **accreted**
    - This means attractive gravitational forces between particles caused them to join together and grow into larger objects

- As the cloud collapsed under gravitational forces
    - it began to spin faster
    - it became hotter
    - it formed an accretion disc

- From the rotating **accretion disc**, the Sun and the planets emerged
    - The Sun formed at the centre
    - The planets formed in the accretion disc

### Formation of the Solar System



### Examiner Tips and Tricks

When writing about the formation of the Solar System, make sure you can:

- State where solid and gaseous matter gathered in the Solar System
- Explain how heavier elements became a part of the nebula
- Describe the role of gravity in pulling matter together and the formation of an accretion disc
- Explain the difference between the planets in terms of the accretion model

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